

COOPER NORMAN

Data Scientist · Production ML & Data Engineering

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Data scientist who takes machine-learning models from data pipeline to production and keeps them reliable.

Solo, I built and ran ShareShark end-to-end, an entire production fintech platform. I owned the full machine-learning system inside it: a calibration-first LightGBM model validated on a 21.5M-row backtest, the leak-safe data pipeline feeding it, and the evaluation, monitoring, and guardrails that kept it reliable in production. I work across the full model lifecycle in Python and SQL. UGA Finance, magna cum laude.

SKILLS

- **Machine Learning:** LightGBM / XGBoost · scikit-learn · probability calibration (Platt / isotonic, ECE / MCE) · SHAP interpretability · Bayesian tuning (skopt) · model evaluation & backtesting · Monte Carlo
- **Data & Engineering:** Python (pandas / NumPy / SciPy) · SQL / PostgreSQL (production) · data pipelines & feature engineering · large-dataset processing · leak-safe / point-in-time validation · data-quality checks
- **Production & Tooling:** model deployment & iteration · performance & data-quality monitoring · automated model testing / guardrails · reproducible workflows (Git) · REST APIs · Celery / Redis · AWS EC2

EXPERIENCE

Founder & Data Scientist — ShareShark

Atlanta, GA / Remote · Live demo: shareshark.app · May 2024 – Present

- Designed and shipped a calibration-first probabilistic ML model (LightGBM), validated on a 21.5M-row temporal-split backtest, with calibration error ~19x lower than a standard baseline (ECE 0.0021 vs 0.0403) and automated testing (Bayesian tuning, calibration metrics) plus auto-reject guards that block bad models before deployment.
- Built the leak-safe data pipeline feeding the models: eight years of data from 6+ vendors, deduplicated and point-in-time-aligned so models never see the future, with automated data-quality validation across sources.
- Deployed behind a REST API on AWS EC2 (Linux), with a staging environment validated before each promotion to production, plus monitoring and a safe fallback.

Independent Quantitative Researcher — Prediction Markets

Nov 2023 – Present

- Build specialized machine-learning models on real-time data to find and hold an edge across skill-based prediction contests and prediction markets, with a fully tracked record of 12,000+ entries at ~9% average ROI and disciplined, risk-based position sizing.

SELECTED PROJECTS

- **Calibrated forecasting models** — A player-outcome system outputting full probability distributions (multiclass PMF) over 1.2M+ player-matches across 37 leagues, with hierarchical support-gated calibration and a leak-safe minutes model. Plus a 9-model pipeline (from-scratch Elo and Glicko-1 ratings) where I caught a dtype bug silently dropping 74% of training data, recovering 10k+ rows (5.2x).

EDUCATION

University of Georgia — B.B.A., Finance (Area of Emphasis: Pricing & Valuation)

May 2024

- Magna cum laude · **GPA 3.83 / 4.00** · HOPE Scholarship · Presidential Scholar · Dean's List